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Energy-level plotting: user guide

This guide explains how to prepare input files for `EnergyLevels.py`, plot energy-level diagrams, and diagnose input errors. It covers global figure settings, dataset definitions, state properties, labels, widths, uncertainties, and links between states. All numerical examples are illustrative.

Run from the extracted code folder. Python 3.10 or newer and Matplotlib are required. All supplied Python files must stay together. SciencePlots is optional. PDF and PNG work without external TeX by default; PGF requires XeLaTeX.

```
python3 -m pip install matplotlib
python3 EnergyLevels.py data.in --check
python3 EnergyLevels.py data.in
```

The check reports the number of datasets and states. It creates no figure and does not import Matplotlib. A successful export prints the saved path. Errors return exit status 2.

Minimal readable input

```
[figure]
output = example
format = pdf
reference_energy = 0

[dataset 1]
name = Experiment
color = -1
linestyle = -

[levels 1]
# id energy width uncertainty j parity label_index jpi bottom top
100  0.000  -1  -1  3/2  +  -1  left  None  None
101  0.467  -1  -1  1/2  +  -1  left  None  None
```

Structure and parsing rules

- One `[figure]` section; each `[dataset N]` needs a matching `[levels N]`. Dataset numbers are positive integers and determine column order. Gaps are allowed.
- Counts are inferred. Add or remove state rows without updating a separate count.
- Blank lines and full lines beginning with `##` are ignored. Do not append inline comments to settings or state rows. Keys and section names are case-sensitive.
- Settings use `key = value`. Their order is free; omitted settings use the defaults on page 2. Do not quote scalar settings or dataset names.
- State rows use whitespace-separated fields. Quote a top/bottom label if it contains spaces. LaTeX backslashes are preserved.

02 / FIGURE

Global settings

All settings below belong in `[figure]`. Energies and energy-axis spacing use MeV. The energy reference shifts every state in every dataset by the same amount.

Setting	Default	Meaning / allowed values
`output`	`energy_levels`	Output stem, without extension. May include an existing directory.
`format`	`pdf`	`pdf`, `png`, `jpg`, `jpeg`, `svg`, `eps`, `ps`, `pgf`.
`dpi`	350	Positive resolution in pixels per inch.
`size_px`	1200, 900	Positive nominal width and height in pixels, separated by a comma.
`ticks`	0.02, 0.2	Minor, major energy-axis tick spacing. Both positive; minor must not exceed major.
`x_margin`	0.5	Extra horizontal padding factor, from 0 to 1. Larger values give more room near the sides.
`y_margin`	0.05	Vertical padding factor, from 0 to 1. Padding is based on the largest absolute plotted energy, with a 1 MeV lower scale.
`width_alpha`	0.4	Width-box opacity: 0 is transparent, 1 is opaque.
`width_spacing`	0.1	Inset at each horizontal end of a width box. Must satisfy $0 \leq f < 0.5$. Box width is $(1 - 2f)$ times level-line length.
`reference_energy`	0	Displayed energy is `energy - reference_energy`. No absolute value or sign reversal is applied.
`ylabel`	Energy (MeV)	Text on the energy axis; mathematical text is allowed.
`font_size`	10	Positive font size in points for ticks and state labels.
`title_size`	11	Positive font size in points for dataset titles and axis label.

Energy-reference example. With input energy −43.145 MeV and reference energy −45.145 MeV, the plot shows 2.000 MeV. With reference zero, the plot shows the input value directly.

Size and resolution. At 350 dpi, a 1200 by 900 input corresponds to approximately 3.43 by 2.57 inches. Font sizes remain in points. Increasing dpi while keeping `size_px` fixed makes the physical figure smaller, so labels occupy a larger fraction of it. PDF/PGF retain vector lines and text.

Paths. Relative input and output paths are resolved from the working directory where you run the command, rather than from the input file's folder. The output directory must exist.

Choosing an output format. PDF is suitable for vector figures in documents; PNG is convenient for image previews. PGF exports drawing commands for inclusion in a LaTeX document and requires XeLaTeX during export. Select the format in the input or override it with `--format`.

03 / DATASETS AND STATES

What each column means

`[dataset N]` supplies defaults for its levels: `name` is the column title (empty means no title), `color` defaults to −1, and `linestyle` defaults to `-`. Valid styles are `-`, `--`, `:`, and `-.`.

Each `[levels N]` row has **ten required fields**, followed by optional settings:

```
id energy width uncertainty j parity label_index jpi bottom top [options]
```

No.	Field	Meaning
1	`id`	Unique, nonnegative integer across the entire file. Used by links and debug labels. IDs need not be consecutive or equal row numbers.
2	`energy`	Finite energy in MeV, before the global reference is subtracted.
3	`width`	Positive: draw a width box from $E - w$ to $E + w$. Use 0 or −1 to disable. See the width convention below.
4	`uncertainty`	Positive: draw an energy error bar from $E - u$ to $E + u$. Use 0 or −1 to disable.
5	`j`	Nonnegative integer or half-integer, e.g. `0`, `2`, `3/2`. Use `-1` only if `jpi` is `None`.
6	`parity`	`+` or `1` for positive; `-` or `0` for negative parity.
7	`label_index`	State-label subscript, e.g. 1 produces J_1^π. Use −1 to omit it. Integer values ≥ -1 are accepted.
8	`jpi`	Physical side of the spin-parity label: `left`, `right`, or `None`.
9	`bottom`	Text below the level, or `None`. Quote text containing spaces.
10	`top`	Text above the level, or `None`. Special value `Energy` displays the shifted energy numerically.

Width convention. The input width is a *half-height for the drawing*: the total box height is $2w$. If a reported physical width Γ is a full width and you want a box of height Γ, enter $w = \Gamma/2$. The program does not perform this conversion automatically. An uncertainty u retains its usual symmetric $\pm u$ meaning.

Labels with spaces or mathematical text

```
100  2.0  0.10  0.03  3/2  -  1  right  "first resonance"  "$\Lambda$"
```

This shows a $3/2_1^-$ label, a 0.20 MeV-high width box and an error bar of ±0.03 MeV. `None` and an empty quoted label both suppress a text label. Do not place both quote types inside the same label; automatic conversion does not support that case.

04 / OPTIONAL CONTROLS

Colors, links and close doublets

Options follow the ten mandatory state fields, in any order. Each option may appear once per row. State options override the dataset defaults.

Option	Default	Meaning
`color=2`	Dataset color	Color for this level and its width/error decorations; integer −1 to 9.
`linestyle=--`	Dataset style	Line style for this level.
`link=100`	No link	Connect this level to the state whose input ID is 100. Any existing ID is allowed except its own.
`link_color=-1`	−1	Link color, integer −1 to 9.
`link_style=--`	`--`	Link style. Same four styles as level lines.
`link_width=0.75`	0.75	Positive link thickness in points.
`label_dy=0.02`	0	Vertical offset of the spin-parity label in MeV. Does not move the level line, energy, width box, error bar or link endpoint.

Color indices (Matplotlib tab10)

−1	Black	0	Blue	1	Orange
2	Green	3	Red	4	Purple
5	Brown	6	Pink	7	Gray
8	Olive	9	Cyan		

A link uses the state ID, not the state-label subscript.

```
[dataset 1]
name = Experiment
[levels 1]
100  0.467  -1  -1  1/2  +  -1  left  None  None

[dataset 2]
name = Calculation
color = 0
[levels 2]
900  0.483  -1  -1  1/2  +  -1  right  None  None  link=100  link_color=-1  link_style=--  link_width
    =0.75
```

Links are drawn from the right end of the column on the left to the left end of the column on the right, independent of ID ordering. Link styling without a `link` destination is rejected. One destination is supported per state row.

Close states. Apply `label_dy=-0.025` and `label_dy=0.025` to separate two nearby labels. Choose offsets small enough to keep the assignment clear. Level lines stay at their true energies. Use `None` in the `jpi` field to suppress an unneeded label. No automatic collision avoidance is applied.

05 / DIAGNOSTICS

Find and fix an input problem

Diagnostic shortcuts

Command / flag	Purpose
`--help`	Show command-line help.
`--check` or `-c`	Check structure, values and references without plotting. `--check` is equivalent.
`--show-ids`	Add the actual input IDs to the diagram in red. Exports a normal figure with these diagnostic labels.
`--debug`	Include a full Python traceback when an error occurs.
`--format png -o preview.png`	Override format and exact output filename. The filename alone does not select the format.
`--tex no`	Use Matplotlib mathtext for labels (default). PGF still needs XeLaTeX.
`--tex yes`	Use external LaTeX for label rendering; requires `latex` and supporting packages.
`--force`	Replace an existing output or converted input.
`--show`	Open an interactive window after saving, if a graphical backend is available.

What an error tells you

```
ERROR: data.in:27 [levels 1] energy:
expected a finite number, got 'wrong'
  > 0 wrong -1 -1 3/2 + -1 left None None
```

Open the reported file at line 27. The section identifies the dataset table; the field identifies which value to fix. The offending row is included. The actual line number depends on the input you edit.

Common fixes

- **Missing columns:** supply all ten required fields, including `None` placeholders for absent top/bottom labels.
- **Duplicate ID:** give each state its own ID, then update links that should target it.
- **Unknown link:** change `link` to an existing input ID. Use `--show-ids` to inspect the plotted assignments.
- **Invalid setting/option:** check spelling, value range and capitalization. Unknown names are rejected rather than silently ignored.
- **TeX/export error:** try `--format pdf --tex no`. For PGF, ensure XeLaTeX is installed. For custom TeX commands, use an appropriate TeX setup or simplify the label.
- **Crowded labels:** increase the physical figure size, reduce font sizes, or apply `label_dy`. Very large offsets can exceed the plot limits; increase `y_margin` if needed.

The checker stops at the first error. It validates structure and numeric values; an actual export is still needed to validate mathematical label rendering and available TeX packages. A rendering error can leave an incomplete output file; fix the cause and use `--force` to retry.

06 / MIGRATION

Use existing inputs and verify the result

Existing files continue to work. The first noncomment line determines the format. A section header such as `[figure]` selects the readable format; `OUTPUT FILE` selects the original format.

```
python3 EnergyLevels.py data_legacy.in --check
python3 EnergyLevels.py data_legacy.in --convert readable.in
python3 EnergyLevels.py readable.in --format pdf -o result.pdf
```

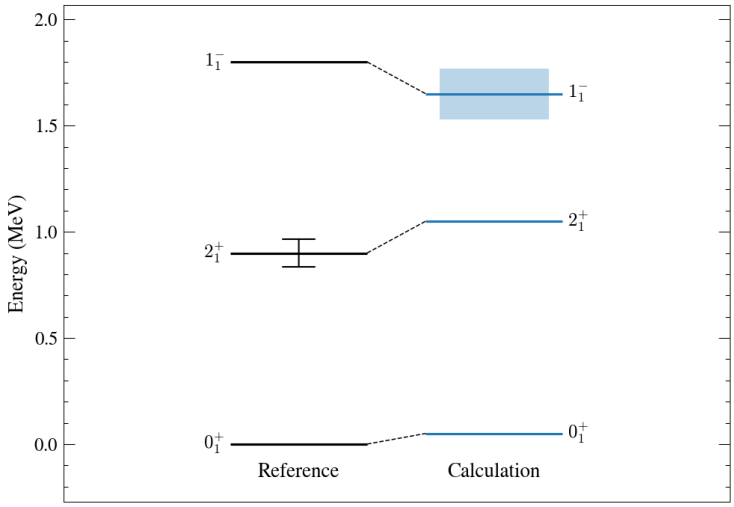
The original source remains intact. Conversion needs a different destination path and refuses to overwrite an existing destination unless `--force` is provided. The original format still requires correct `NUM DATA` and per-dataset state counts, but blank lines and full-line comments no longer break row offsets.

Old option names remain accepted

Original syntax	Readable equivalent
`C##2`	`color=2`
`LS##--`	`linestyle=--`
`L##100`	`link=100`
`LC##-1`	`link_color=-1`
`LLS##--`	`link_style=--`
`LLW##0.5`	`link_width=0.5`

The original renderer named the physical left and right text positions in reverse. The legacy reader translates those values to preserve existing label positions. Readable inputs use the actual physical side. Conversion preserves energies, widths, uncertainties, IDs and link assignments.

Illustrative diagram



Two datasets compare illustrative reference and calculated levels. Dashed links identify corresponding states; a shaded box shows the input width convention, and a capped error bar represents the uncertainty. The complete input is provided in `general_example.in`.

```
python3 EnergyLevels.py general_example.in --check
python3 EnergyLevels.py general_example.in -o example.pdf
```

Before sharing a figure. Confirm the energy reference and units, check state IDs and link assignments, inspect label placement, and verify that the width and uncertainty conventions match the intended physical quantities.